

REFLECTION JOURNAL: MODEL EVALUATION

Course: ITAI1371

Lab Assignment: Lab 07

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Date: July 22, 2026

1. Key Insights and Concept Mastery

The primary takeaway from Lab 07 is that standard classification accuracy is a highly deceptive metric that frequently hides model flaws. Initially, our Logistic Regression model scored an overall test accuracy of 80.45% on the Titanic dataset, which superficially implies a highly successful predictive system. However, breaking this down using a Confusion Matrix exposed a critical performance asymmetry between the passenger classes.

When evaluating the survival class, the model achieved a Precision of 0.78 and a Recall of 0.73. This mathematically proves that while the model is relatively dependable when it flags a survivor (78.3% correct), it suffered from 20 False Negatives—meaning it completely missed 27% of the actual survivors by misclassifying them as deceased. Learning to evaluate models through class-specific precision and recall metrics is an essential engineering skill for catching these hidden algorithmic weaknesses before code is ever deployed to a production environment.

2. Analytical Challenges and Technical Breakthroughs

Our primary technical hurdle during this laboratory assignment was identifying and overcoming the random sampling bias inherent in a single standard train-test split. A single 80/20 data partition can easily create a "lucky" or "unlucky" data slice that overstates or understates model capabilities. For instance, our initial baseline cross-validation loops hit a high score of 80.34% on an easy data chunk, while dropping down to 77.53% on a more difficult chunk, resulting in a baseline average of 78.68%.

To solve this rigorously, I expanded our workflow beyond the lab's basic requirements by implementing a 25-Run Repeated Cross-Validation strategy. This process reshuffled the entire

dataset randomly across 5 separate shuffles, generating 25 unique evaluation loops. This technique completely smoothed out the localized sampling spikes, yielding an authentic, unbiased grand average accuracy of 78.76%.

Furthermore, I applied probability theory to calculate a 95% Confidence Interval of [77.75%, 79.78%]. This statistical addition was a breakthrough for our team. Instead of guessing how our model will perform on unseen real-world data, we now have a mathematical guarantee of its real-world performance boundaries.

3. Real-World Connections and Engineering Takeaways

This lab bridges the gap between academic exercises and production-grade machine learning engineering. In real-world business scenarios, default classification thresholds cannot be left untouched because different errors carry different business weights. If we were building an automated fraud prevention system, a False Positive (freezing an innocent user's account) carries an incredibly high operational cost and damages customer retention, meaning we would manually tune the model for High Precision. Conversely, in an airport security scanner looking for weapons, a False Negative (missing a weapon) is catastrophic, forcing us to optimize High Recall at the expense of generating a high volume of false alarms.

Ultimately, this assignment taught me that selecting evaluation metrics is not just a mathematical exercise, it is a critical business, safety, and design decision that directly impacts user experience, operational overhead, and overall software system reliability.
